Credit Risk Classification Report

Introduction

This project investigated the use of machine learning for credit risk prediction using the German Credit dataset. The main objective was to compare Logistic Regression and Random Forest in classifying applicants as either low risk or high risk. This kind of analysis is valuable to financial institutions because it supports better lending decisions and helps reduce loan default losses.

Data Understanding

The dataset contained demographic, financial, and personal information about loan applicants, including age, sex, job type, housing status, savings accounts, checking accounts, credit amount, duration, and purpose. The target variable was `Risk`, which indicated whether an applicant was classified as good or bad credit risk. The dataset was inspected to understand its structure, identify missing values, and confirm the presence of the target column.

Data Preparation

Data cleaning was performed to improve the quality of the dataset before modeling. Missing values in categorical variables were replaced with the most frequent category, while missing numerical values were replaced with the median. The target column was encoded into binary form, and categorical predictors were transformed using one-hot encoding. This ensured that the dataset was fully compatible with machine learning algorithms.

Exploratory Data Analysis

Exploratory analysis was conducted to examine the distribution of credit risk, age, and credit amount. The visualizations helped reveal how applicants were spread across the two risk categories and showed the distribution of key financial variables. A boxplot of credit amount against risk was used to explore whether loan size differed between good and bad risk applicants.

Model Development

Two supervised classification models were built for this study: Logistic Regression and Random Forest. Logistic Regression was trained on standardized data because it is sensitive to feature scale. Random Forest was trained on the encoded dataset without scaling, as tree-based models are not affected by feature magnitude. Both models were trained on the same train-test split to ensure a fair comparison.

Model Evaluation

The models were evaluated using accuracy, precision, recall, and F1 score. Accuracy measured the proportion of correct predictions overall. Precision indicated how many predicted risky applicants were actually risky, while recall measured how well the model identified all risky applicants. F1 score provided a balanced measure of precision and recall. A comparison table and confusion matrices were used to summarize the results clearly.

Recommendation

The preferred model is the one with the strongest balance of evaluation metrics, especially F1 score and recall. In a credit risk context, recall is especially important because failing to identify high-risk applicants can lead to financial losses. For this reason, the better-performing model should be recommended for deployment.

Conclusion

This project demonstrated that machine learning can be effectively applied to credit risk assessment. By comparing Logistic Regression and Random Forest, the analysis showed how different algorithms perform on the same problem. The final recommendation should be based on the model that offers the best trade-off between predictive reliability and business risk.